Q1 What is the value of capacitance, if a very thin metallic plate (thickness t is introduced between two parallel plates of area A and separated at distance d?

(A) $\frac{\varepsilon_0 A}{d-t}$

(B) $\frac{\varepsilon_0 A}{d}$

(C) $\frac{2\varepsilon_0 A}{d}$

(D) $\frac{3\varepsilon_0 A}{2d}$

Q2 The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

(A) Proportional to the square root of the distance between the plates

(B) Linearly proportional to the distance between the plates

(C) Independent of the distance between the plates

(D) Inversely proportional to the distance between the plates

Q3 The capacitance of a parallel plate capacitor is $60\mu\text{F}$. If the distance between the plates is tripled and area doubled then new capacitance will be

(A) $10\mu\text{F}$

(B) $40\mu\text{F}$

(C) $80\mu\text{F}$

(D) $100\mu\text{F}$

Q4 The potential of the two plates of capacitor are $+10\text{ V}$ and -10 V. The charge on one of the plates is 40C. The capacitance of the capacitor is

(A) 2 F

(B) 4 F

(C) 0.5 F

(D) 0.25 F

Q5 The charge stored in a capacitor is $20\mu\text{C}$ and the potential difference across the plates is 500 V. Its capacity is

(A) $0.04\mu\text{F}$

(B) $10^{-2}\mu\text{F}$

(C) $2\times10^{-6}\mu\text{F}$

(D) $250\mu\text{F}$

Q6 A solid conducting sphere of radius R_1 is surrounded by another concentric hollow conducting sphere of radius R_2. The capacitance of this assembly is proportional to:

(A) $\frac{R_2-R_1}{R_1R_2}$

(B) $\frac{R_2+R_1}{R_1R_2}$

(C) $\frac{R_1R_2}{R_1+R_2}$

(D) $\frac{R_1R_2}{R_2-R_1}$

Q7 The capacitance C of a capacitor is

(A) independent of the charge and potential of the capacitor

(B) dependent on the charge and independent of potential

(C) independent of the geometrical configuration of the capacitor

(D) independent of the dielectric medium between the two conducting surface of the capacitor

Q8 A capacitor is said to be fully charged when:

(A) Current in the circuit becomes zero

(B) Potential difference across plates becomes zero

(C) Potential of both plates becomes equal

(D) Charge on both plates becomes zero

Q9 Which of the following correctly defines a capacitor?

(A) A device that stores magnetic energy in the form of current.
(B) A device that stores electric charge and potential energy.
(C) A device that converts electrical energy into mechanical energy.
(D) A resistor-like device that opposes current flow.

Q10 What physical quantity is stored in a capacitor, and what are its units?
(A) Energy, joule
(B) Charge, coulomb
(C) Electric field, V/m
(D) Current, ampere

Answer Key

Q1	A
Q2	C
Q3	B
Q4	A
Q5	A
Q6	D
Q7	A
Q8	A
Q9	B
Q10	A

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

A thin inserted metallic plate becomes an equipotential surface, so the electric field (and hence potential drop) exists only across the remaining air gap of width $d - t$. For a parallel-plate capacitor in vacuum/air,

$$C = \frac{\varepsilon_0 A}{\text{separation}} = \boxed{\frac{\varepsilon_0 A}{d - t}}.$$

Video Solution:

Q2 Text Solution:

1. **Field between the plates (isolated capacitor, charge Q fixed)**
   Surface charge density $\sigma = Q/A$.
   Uniform field $E = \sigma/\varepsilon_0 = Q/(\varepsilon_0 A)$ between the plates .
2. **Stored energy**
   The energy in a parallel-plate capacitor is

   $$U = \tfrac{1}{2}\,\varepsilon_0 E^2 A d$$

   where d is the plate spacing and A the plate area .
3. **Express U in terms of Q**
   Substitute $E = Q/(\varepsilon_0 A)$:

   $$U = \tfrac{1}{2}\,\varepsilon_0 \left(\frac{Q}{\varepsilon_0 A}\right)^2 A d = \frac{Q^2 d}{2\varepsilon_0 A}.$$

4. **Force from energy gradient**
   For fixed charge, the attractive force equals the negative gradient of energy with respect to separation:

   $$F = -\frac{dU}{dd} = \frac{Q^2}{2\varepsilon_0 A}.$$

   The expression contains **no d**, so the force does **not change** when the plates are moved closer or farther apart.

Hence, the electrostatic force between the charged plates of an isolated parallel-plate capacitor is **independent of the distance between them**.

Video Solution:

Q3 Text Solution:

$\frac{\varepsilon_0 A}{d} = 60\mu\text{F}$ ...(1)

$\frac{2\varepsilon_0 A}{3d} = C'$ ...(2)

on deviding (1) / (2)

we get

$\frac{60}{C'} = \frac{3}{2}$

$C' = 60 \times \frac{2}{3} = 40\mu\text{F}$

Video Solution:

Q4 Text Solution:

Answer: A ($2\ \text{F}$)

$Q = C\,\Delta V$

$C = \frac{Q}{\Delta V}$

$= \frac{40}{20}$

$= 2F$

Video Solution:

Q5 **Text Solution:**

$C = \frac{q}{V}$

$q = 20 \times 10^{-6}\ C \qquad v = 500v \qquad c$

$=?$

$c = \frac{20\times10^{-6}}{500}$

$= 0.04 \times 10^{-6}\ \ F$

Video Solution:

Q6 **Text Solution:**

$$C \propto \frac{R_1 R_2}{R_2 - R_1}$$

Key steps (concise)

1. Treat the two concentric conductors as a spherical capacitor (inner radius R_1, outer R_2).
2. Potential difference
$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$
3. Capacitance $C = \frac{q}{V} = \frac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1}$.

Hence C is proportional to $\frac{R_1 R_2}{R_2 - R_1}$.

Video Solution:

Q7 **Text Solution:**

Reason (concise)

$C = Q/V$; this ratio is fixed for a given geometry and dielectric, so C does **not** vary with how much charge you place or what potential difference you apply. It **does** depend on size/shape and the medium between the plates (ε).

Electrostatic Potential and Capacitance" (section on definition of capacitance).

Video Solution:

Q8 **Text Solution:**

In DC circuits, a capacitor is fully charged when no more current flows, i.e., current becomes zero and voltage across plates equals the source EMF.

Video Solution:

Q9 **Text Solution:**

A capacitor is used to store electric charge and potential energy in the electric field between its plates.

Video Solution:

Q10 **Text Solution:**

A capacitor stores electric potential energy, not charge itself. Unit of energy is joule.

Video Solution:

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